

ASSESSMENT-1

Course Name : NLP
DEADLINE : 04.10.2026

Domain-Specific Text Analysis and Retrieval System

1. Objective

Develop an NLP-based domain-specific text analysis and information retrieval system using the concepts covered in the course.

The main focus is to:

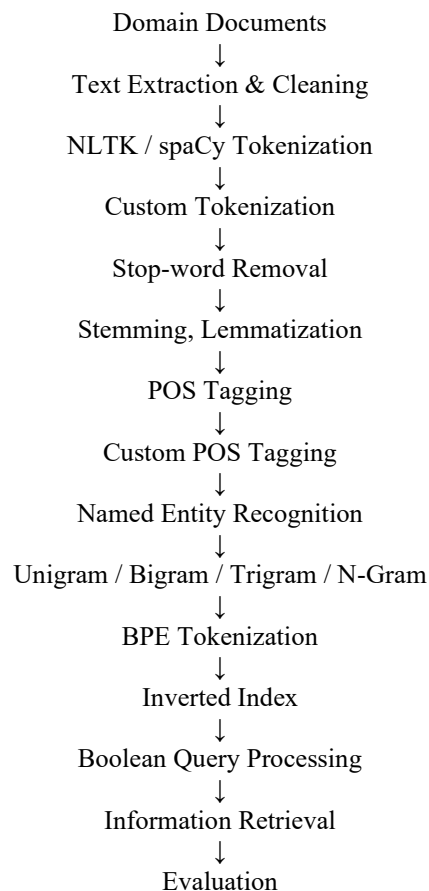
Select → Order → Implement → Compare → Evaluate → Justify

NOTE: Students must understand not only how to apply NLP techniques, but also why a particular process is selected, why it is placed at a particular stage, and how changing the process/order affects the results.

2. Domain and Dataset

- Consider an application domain which is being selected as your NLP project domain.
- Collect a minimum of 15 domain-specific documents. **Note:** Recommended: 15–30 documents.
- Supported formats: PDF, TXT, CSV, HTML, JSON, DOCX, etc.
- Assign document IDs such as D01, D02, ... D30

3. Required NLP Pipeline



Note: Students may modify this pipeline based on their domain/ problem requirement, but **the selection and order must be justified.**

4. Project Modules

Module 1 – Document Collection and Extraction

- Collect 15+ domain-specific documents.
- Load documents from Google Drive/local storage.
- Extract and clean text.
- Create a unified corpus.
- Generate document statistics.

Required statistics:

- Number of documents
- Number of sentences
- Number of tokens
- Number of characters
- Vocabulary size
- Average document length

Module 2 – Preprocessing

Exercise 1. Observe the size of the dictionary and compare with term-based dictionary.

Exercise 2. Observe the tokens in the dictionary and identify the processes that can be done to improve the performance after applying different type of built-in tokenization methods.

Exercise 3. Custom tokenization:

Identify at least five domain-specific tokenization problems.

Examples:

AI-based

COVID-19

B.Tech

₹50,000

machine-learning

Develop at least one custom tokenizer rule.

- Compare *NLTK* vs *spaCy* vs *Custom Tokenizer* vs *Hybrid Tokenizer* and explain whether the custom tokenizer improves the result. (**Note:** Try with different algorithms and do comparisons with examples and select the suitable method with the proper justification)

Exercise 4. BPE Tokenization

Apply BPE using the method/tool covered in the course.

Compare:

NLTK vs spaCy vs Custom Tokenizer Vs BPE Vs Hybrid

Analyze:

- Common words
- Rare words
- Domain-specific words
- Subword formation
- Vocabulary size
- Token count

Exercise 5: Identify scenario/tasks/application related to your domain where

- i) both verbs and Noun phrase are useful
- ii) Only Noun phrases are useful. But, verbs are not important

Exercise 6: Identify whether your domain require date and numbers or not. Accordingly do tokenization.

Exercise 7: Stop-word removal (Justify your process)

Note: Stop words are words which are filtered prior to or after processing of natural language data (text). Note that stop word removal is not a necessary step for dictionary

building. One has to decide based on the domain/application whether to use it or not. Also there is not one definite list of stop words which can be used across the tools and /or applications. Specific stop word lists can be generated according to the need of the application.

Exercise 8: Stemming (Note: Try with different algorithms and do comparisons with examples and select the suitable method with the proper justification. Focus on over-stemming and under-stemming.)

Exercise 9: Analyze the importance of order of stemming and stop-words removal process.

Exercise 10: Lemmatization (Note: Try with different algorithms and do comparisons with examples and select the suitable method with the proper justification)

Exercise 11: Compare the corpus before and after preprocessing.

Exercise 12: POS Tagging and Custom POS Tagger

- Apply POS tagging using NLTK/spaCy.
- Identify at least **5 domain-specific terms** that may be incorrectly tagged.
- Develop a simple custom POS tagger using domain-specific rules/dictionaries.
- Develop a simple custom POS tagger using ML algorithm.
- Compare default and custom POS tagging.
- Calculate accuracy/improvement where possible.

Exercise 13: – Named Entity Recognition

Apply NER and identify entities such as:

- PERSON
- ORG
- GPE
- DATE
- MONEY
- PRODUCT
- EVENT

Analyze:

- Correct entities
- Incorrect entities
- Missed entities
- Domain-specific entities

Exercise 14: – N-Gram Analysis

Generate:

- Unigram
- Bigram
- Trigram
- 4-Gram
- 5-Gram

For each, calculate:

- Total count
- Unique count
- Frequency
- Top 10 N-grams

Note: Identify meaningful domain-specific phrases.

Also, Observe the size of the dictionary and compare with term-based dictionary.

Module 3. Process Selection and Ordering – Mandatory

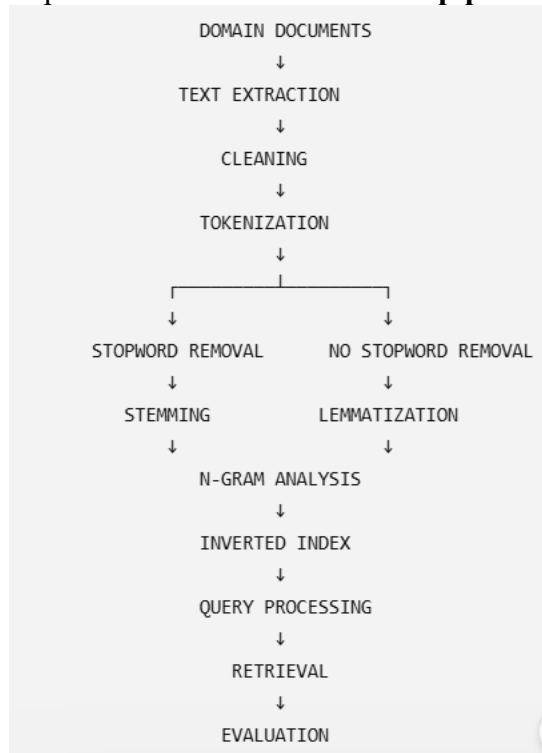
Students must **design and justify their NLP pipeline.**

For each process, explain:

- Why it is required/not required.
- Where it is placed.
- What dependency it has on other processes.
- What happens if its position is changed.

Mandatory Experiment

Implement at least **two different pipeline orders Pipeline A and Pipeline B.**



Compare:

- Token count
- Vocabulary size
- N-Gram quality
- Domain terms

Finally, select the best pipeline and justify the selection.

Module 4. Information Retrieval

Build an **inverted index/posting list**.

Example:

machine → D01, D03, D05

learning → D01, D02, D05

robotics → D02, D04

Implement:

- Keyword search
- Phrase search
- Boolean AND
- Boolean OR
- Boolean NOT

Module 5. Domain-Specific Query Analysis

Create **10–15 domain-specific queries**.

Examples:

machine

machine learning

machine learning model

machine AND learning
machine OR robotics
machine AND NOT robotics

For each query, record:

Query | Type | Retrieved Documents | Number of Results

Module 6. Retrieval Evaluation

Create a small relevance set by identifying relevant documents for each query.

Calculate, where applicable:

- Precision
- Recall
- F1-score
- Precision@K
- Recall@K

Required Results Format

A. Document & Preprocessing Results

Measure	Before Processing	After Processing
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Documents

Tokens

Unique Tokens

Vocabulary Size

B. Tokenization Comparison

Input	NLTK	spaCy	Custom
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Example 1

Example 2

Example 3

C. Stemming vs Lemmatization

Word	Stemmed	Lemmatized
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D. POS Tagging

Word	Default POS	Custom POS	Correct/Incorrect
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E. NER

Entity	Predicted Type	Correct?
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Report total, correct, incorrect and missed entities.

F. N-Grams

N-Gram	Total Unique	Top N-Grams
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Unigram

Bigram

Trigram

4-Gram

5-Gram

G. BPE

S.No.	Word	BPE Tokens
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H. Pipeline Comparison

Measure	Pipeline A	Pipeline B	Final Pipeline
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Token Count

Vocabulary Size

Meaningful N-Grams

Precision

Recall

Measure	Pipeline A	Pipeline B	Final Pipeline
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F1-score

I. Retrieval Results

Query Type	Retrieved Documents # Results Time
Unigram	
Bigram	
Trigram	
Boolean AND	
Boolean OR	
Boolean NOT	

J. Overall Performance

Method/Pipeline	Precision	Recall	F1-score	Time
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Pipeline A

Pipeline B

Final Pipeline

6. GUI – Required Component

Students must develop a **simple GUI/interface** to demonstrate the completed system.

The GUI should provide:

1. Document Upload/Loading
2. Document Statistics
3. Tokenization
4. Preprocessing
5. POS Tagging
6. Custom POS Tagging
7. NER
8. N-Gram Analysis
9. BPE Analysis
10. Inverted Index
11. Query Input
12. Query Type Selection
13. Document Retrieval
14. Pipeline Comparison
15. Performance/Evaluation

Sample Suggested GUI

DOMAIN-SPECIFIC TEXT ANALYSIS & RETRIEVAL

[Load Documents]

Document Statistics

Tokenization

Preprocessing

POS Tagging

Custom POS

NER

N-Gram Analysis

BPE Analysis

Query: _____

Query Type: [Keyword ▼]

[SEARCH]

Retrieved Documents:

D01 D04 D07

Number of Results: _____

Execution Time: _____

Precision: _____

Recall: _____

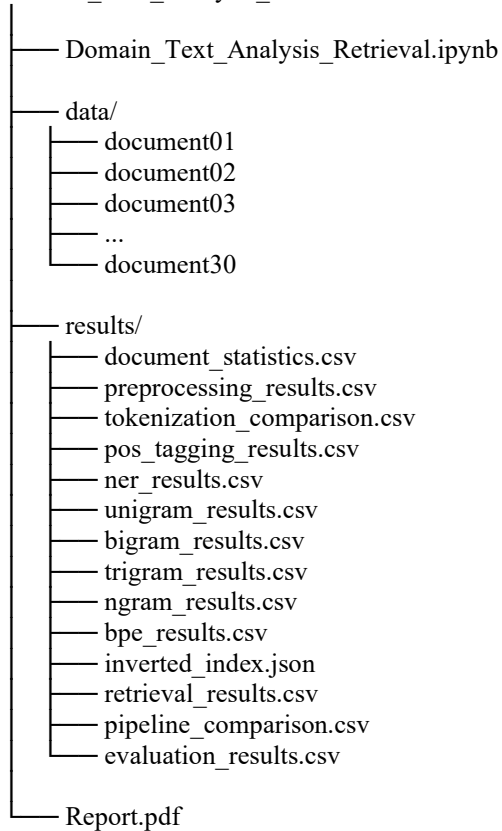
F1-Score: _____

[Compare Pipelines]

Note: The GUI does not need to be complex; it should demonstrate the complete working pipeline.

11. Sample Submission Structure

Domain_Text_Analysis_Retrieval/



Sample GUI:

